Ground shakes: what we found

We see a clear disturbance in the saved detector signals. We still do not know what causes it.

“Ground shake” is the name used for this disturbance. The records measure electrical signals in the detector. They do not measure physical shaking.

The records we checked

We reviewed 1,531 saved records from seven ProtoDUNE-HD runs in July, October and November 2024. An automatic check found 1,516 records with a large disturbance on both charge-readout panels. These selected files cannot tell us how often this happens in normal operation.

The charge detector, or TPC, records signals on wires. The light-detector system, or PDS, records signals through DAPHNE electronics. We compared their signal sizes, times and positions. We also checked archived voltage and current records.

The checks and what they tell us

Early pulses. We found sharp PDS pulses before the main disturbance in 1,179 of the 1,516 selected records. Missing recording time prevents us from saying whether every disturbance has an earlier pulse.

Two pulses. In run 32731, 101 of 716 selected records had at least two pulses between 300 and 100 microseconds before the main disturbance. A microsecond is one millionth of a second. These are pulse candidates, not confirmed flashes of light.

Electrical response. We used inputs reported as disconnected to predict signals on connected inputs. They explained part of the response. This does not separate all electrical interference from real light.

Charge and light timing. We compared pulse times with nearby activity on TPC wires. A delay pattern remained when we swapped PDS records between events. We have not identified a unique charge–light match. A source repeating in the same place remains possible.

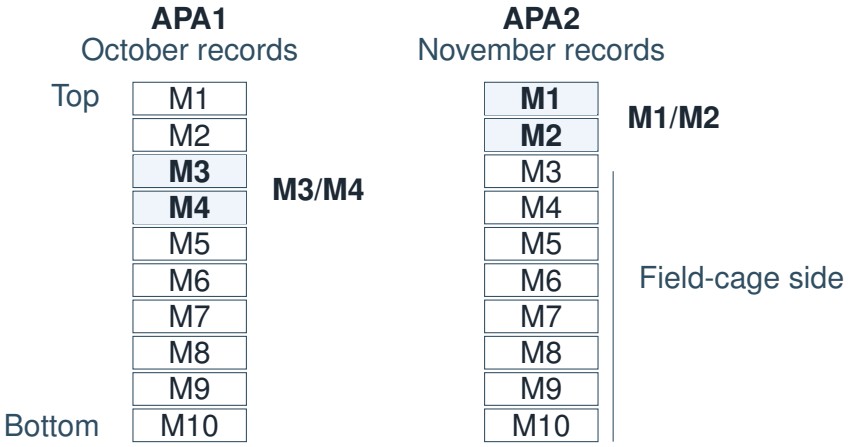
Position and electronics. The slow activity before the main disturbance was strongest in upper modules. Signals sharing an electronics group also tended to match more closely during the main disturbance. Neither result locates the cause.

Voltage and current. The checks found no clear relationship that held across both November runs. These minute-scale records cannot show very fast voltage or current changes.

Reading the detector map

A channel is one recorded input. A module is a physical light-detector unit with four readout channels. The modules sit in the charge-readout panels, called APA1 and APA2. M means module.

Upper modules had the strongest slow activity



Separate views in the source-map orientation; spacing is schematic. Bold rows mark the strongest slow activity in 39 of 49 usable APA1 records and 1,210 of 1,224 APA2 records. These are different runs, not simultaneous views of the same events.

The channel orientation was checked

We checked all 40 channel positions in each map against the historical source. M1 is at the top; M10 is at the bottom. In the APA2 view, the field cage is on the right. The field cage shapes the electric field inside the TPC. The top row reads **27, 25, 22, 20** from left to right. Channel 20 is at the top-right corner.

The source photograph was mirrored to match this view. APA1 has a different channel order. This is a check against the historical map, not a new survey of the installed detector. We have not assigned beam or compass directions.

What we still need to learn

We have not found one module that always starts first. Several modules often start too close together to separate with the current timing. We need longer recordings. We also need to measure each channel's own delay and use the wire signals to work out where charge came from. These checks can help test where the disturbance begins and whether light is involved.

Sources. A. Cervera, EDMS 3493627 v1, pp. 18, 20–21 (historical working presentation). Module measurements, timing checks, and full source notes accompany this report.